

|              | AGENT ON EVERY MACHINE                                                                                        | WHEN IT RUNS         | CREATES THE MACHINES  |               |
|--------------|---------------------------------------------------------------------------------------------------------------|----------------------|-----------------------|---------------|
|              | None, over SSH                                                                                                | When we run it       | Yes, same tool        | <b>CHOSEN</b> |
|              | A push over SSH puts nothing new on a CERN host. One tool creates the five machines and configures them.      |                      |                       |               |
| with Foreman | Agent and certificate                                                                                         | On a timer           | Yes, with Foreman     |               |
|              | The CERN standard, and it repairs drift by itself. It wants an agent and a signed certificate on every node.  |                      |                       |               |
| and Chef     | Agent and a server                                                                                            | On demand or a timer | No                    |               |
|              | Salt fans out fastest, but only at farm scale. Chef needs a second language. Both put an agent on every node. |                      |                       |               |
|              | None                                                                                                          | When we run it       | Yes, and nothing else |               |
|              | It builds machines well, and its vendor calls configuring them a separate job. That is two tools, not one.    |                      |                       |               |

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|--|-------------------------------|-----------------------------------------------------------------------------------------------------------|
|  | <b>Placement is data</b>      | One inventory file says where each service runs. Moving one service to another machine changed no code.   |
|  | <b>Every step converges</b>   | A failed run is simply run again. Machines, monitors and detectors are created by name, never twice.      |
|  | <b>Gates stop the run</b>     | 12 checks fail the deploy rather than report success. The last proves 28 monitors and 17 detectors exist. |
|  | <b>Secrets stay encrypted</b> | Every secret lives in one encrypted file in the repository. The cloud credential never touches disk.      |